GPS-Based Travel Companion - Real-Time Navigation and Trip Monitoring System

Arduino Nano + GPS Module + 16×2 Parallel LCD

1. Abstract

This project implements a GPS-based travel monitoring system using an Arduino Nano, a GPS receiver and a 16×2 parallel LCD. The program receives GPS NMEA data through SoftwareSerial and processes it using the TinyGPS++ library. The system displays vehicle speed, satellite count, latitude, longitude, altitude, direction, Indian Standard Time (IST), date, distance travelled, trip time, maximum speed, average speed, moving/stopped status, GPS status, heading and GPS fix age.

The design does not require additional sensors for the listed functions. The GPS receiver supplies position, speed, course, altitude and UTC date/time information. The Arduino processes these values and presents them through a sequence of LCD pages. Serial Monitor output is also provided every second for testing and data observation.

2. Objectives

- Measure and display the current vehicle speed using GPS data.
- Monitor latitude, longitude and altitude.
- Determine direction of travel from GPS course information.
- Calculate travelled distance and trip duration.
- Record maximum, minimum and average speed values.
- Detect whether the vehicle is moving or stopped.
- Convert GPS UTC time to IST, including date rollover.
- Provide GPS status, satellite count, heading and fix-age information.

3. System Overview

The GPS module communicates with the Arduino through two serial pins. The Arduino decodes the incoming GPS data and updates the calculated travel parameters. The 16×2 LCD is connected in parallel mode and is used as the primary display. The display automatically changes page every 10 seconds, while the Serial Monitor is refreshed every second.

4. Main Hardware

Component	Purpose
Arduino nano	Main controller and data processing
GPS module	Position, speed, course, altitude and UTC time/date
16×2 LCD	Local display of calculated and GPS data
Connecting wires	Electrical connections between modules

Hardware Connections

5. Pin Configuration

The code uses the following connections. These connections match the pin definitions in the supplied Arduino program.

Device	Pin	Arduino Pin	Function
GPS	TX	D7	GPS data into Arduino (SoftwareSerial RX)
GPS	RX	D6	Arduino data to GPS (SoftwareSerial TX)
GPS	VCC	5V	Power supply
GPS	GND	GND	Common ground
LCD	RS	D2	Register Select
LCD	EN	D4	Enable
LCD	D4	D9	LCD data bit 4
LCD	D5	D10	LCD data bit 5
LCD	D6	D11	LCD data bit 6
LCD	D7	D12	LCD data bit 7
LCD	VSS/GND	GND	Ground
LCD	VCC	5V	Power supply

6. Software Configuration

Parameter	Value
GPS RXPin	7
GPS TXPin	6
GPS baud rate	9600
LCD interface	16×2 parallel
LCD constructor	LiquidCrystal(2, 4, 9, 10, 11, 12)
LCD page interval	10 seconds
Serial output interval	1 second
Moving threshold	2.0 km/h

7. Required Arduino Libraries

- SoftwareSerial – creates a serial interface for the GPS module.
- TinyGPS++ – parses GPS data and provides location, speed, course, altitude, date and time.
- LiquidCrystal – controls the 16×2 parallel LCD.
- math.h – included for mathematical support.

8. Connection Notes

The GPS TX line is connected to Arduino pin D7 because the program defines D7 as the SoftwareSerial receive pin. GPS RX is connected to D6, the SoftwareSerial transmit pin. The LCD uses six Arduino digital pins: D2, D4, D9, D10, D11 and D12. All modules should share a common ground. The exact power requirements of the particular GPS module should be checked before applying 5 V.

9. Signal Flow

GPS satellite signals → GPS receiver → serial NMEA data → Arduino/TinyGPS++ → calculations → 16×2 LCD and Serial Monitor.

Software Design and Working

10. Program Operation

After startup, the Arduino initializes the USB Serial interface, GPS SoftwareSerial interface and LCD. The LCD initially shows “Searching Sat...” and the satellite count. The program then continuously reads available GPS characters and feeds them to TinyGPS++.

11. GPS Lock and Trip Start

The system checks gps.location.isValid(). When a valid position is obtained for the first time, the trip timer starts and the initial latitude and longitude are stored. This position becomes the reference point for subsequent distance calculations.

12. Distance Calculation

When the GPS location is updated, TinyGPSPlus::distanceBetween() calculates the distance between the previous and current coordinates. The returned value is in metres and is converted to kilometres. Movements smaller than 2 metres are ignored to reduce the effect of small GPS position fluctuations.

13. Speed Statistics

The program reads GPS speed in kilometres per hour. Maximum speed is updated whenever a new value is higher than the stored maximum. Minimum speed is updated whenever a new value is lower. A sample average is calculated from speedSum divided by speedSamples.

14. Trip Average Speed

A separate trip-average calculation is included. It uses total distance divided by elapsed trip time in hours. This gives a journey-level average based on distance and elapsed time rather than simply averaging individual GPS speed samples.

15. Moving / Stopped Detection

The program uses a threshold of 2.0 km/h. When valid GPS speed is at least 2.0 km/h, the vehicleMoving flag becomes true and the LCD reports MOVING. Below the threshold, it reports STOPPED. This is a GPS-based indication and may fluctuate when the GPS signal is noisy.

16. Direction and Heading

GPS course is used to determine direction of travel. The numeric heading in degrees is mapped into eight compass directions: N, NE, E, SE, S, SW, W and NW. The system therefore reports direction of movement rather than wind direction.

17. IST Conversion

GPS time is normally UTC. The program adds 5 hours and 30 minutes to obtain IST. It also handles midnight rollover and updates the day, month and year, including February and leap-year rules. Therefore, a UTC time near midnight can correctly produce the following IST calendar date.

LCD Pages and Data Output

18. LCD Display Pages

Page	Displayed Information	Purpose
1	Speed + Satellites	Current speed and satellite availability
2	Latitude + Longitude	Current GPS coordinates
3	Altitude + Direction	Height and direction of travel
4	IST Time + Date	Local Indian time and date
5	Distance + Trip Time	Journey distance and elapsed time
6	Maximum + Sample Average	Speed statistics
7	Moving / Stopped	Vehicle movement status
8	GPS Status + Satellites	GPS lock and satellite count
9	Heading + Fix Age	Direction in degrees and freshness of position
10	Trip Average + Minimum	Journey average and minimum speed

19. Example LCD Screens

Screen	Example
Page 1	Spd: 45.6km/h / Sats: 8
Page 2	Lat:10.1234 / Lng:76.1234
Page 3	Alt:125.4m / Dir:NE
Page 4	10:05:30 IST / 07/10/2026
Page 5	Dist:12.5km / Trip:00:24:15
Page 6	Max:65.2km/h / Avg:42.3km/h
Page 7	Vehicle: / MOVING
Page 8	GPS:OK / Sats:8
Page 9	Head:72° / Fix:1s
Page 10	TAvg:31.0km/h / Min:0.0km/h

20. Serial Monitor

The Serial Monitor is updated every second. It reports GPS status, satellite count, latitude, longitude, speed, vehicle status, maximum speed, minimum speed, sample average speed, trip average speed, altitude, direction, heading, GPS fix age, travelled distance, trip time, GPS update count, UTC time, IST time/date and GPS character/checksum statistics.

21. GPS Fix Age

gps.location.age() is used to indicate how long it has been since the location was updated. A small age generally means that the displayed position is recent. The LCD displays the age in milliseconds when it is below one second and in seconds otherwise.

22. Automatic Page Switching

The LCD page changes every 10 seconds using millis(). This non-blocking timing approach allows the GPS serial data to continue being processed while the display changes. The Serial Monitor uses a separate one-second interval.

Advantages, Limitations and Conclusion

23. Advantages

- Uses GPS data for multiple travel-monitoring functions without additional sensors.
- Provides both LCD and Serial Monitor outputs.
- Automatically cycles through ten information pages.
- Includes distance calculation with a small-position-noise filter.
- Provides both sample average and journey-level trip average speed.
- Includes moving/stopped detection using a configurable speed threshold.
- Converts UTC GPS time to IST with date rollover.
- Provides diagnostic information such as GPS status, fix age and checksum count.

24. Limitations

- GPS speed and position accuracy depend on satellite visibility and signal quality.
- Direction/course is most meaningful while the vehicle is moving; it can be unreliable at very low speed.
- Moving/stopped status can fluctuate if GPS speed fluctuates around the 2 km/h threshold.
- Distance can contain GPS-related error, especially in poor reception conditions.
- Altitude from GPS is generally less precise than horizontal position.
- The system does not measure wind speed because no wind sensor is included.

25. Future Improvements

- Add a reset mechanism for trip statistics if user controls are later permitted.

- Store trip information in memory or an SD card for later analysis.
- Add graphical or smartphone visualization using a suitable communication module.
- Use a more advanced display for simultaneous presentation of more parameters.
- Apply filtering or averaging to reduce GPS noise in speed, position and direction.

26. Conclusion

The developed GPS Travel Monitor is a compact Arduino-based travel information system. The supplied program integrates GPS decoding, travel calculations, time conversion and LCD presentation into a single system. It can display the current speed, coordinates, altitude, direction, time, date, distance, trip duration, speed statistics, vehicle status, GPS status, heading and fix age.

The project demonstrates how a microcontroller can transform raw GPS information into practical vehicle-travel data. Because the system is based on the GPS receiver and the existing LCD, the major monitoring functions can be implemented without adding separate speed, distance or direction sensors. The design is therefore suitable as an educational embedded-systems project and as a foundation for a more advanced travel-monitoring device.

27. Key Technical Summary

Item	Specification
Controller	Arduino Nano
GPS Interface	SoftwareSerial, 9600 baud
GPS RX/TX	D7 / D6
Display	16×2 parallel LCD
LCD Pins	D2, D4, D9, D10, D11, D12
Page interval	10 seconds
Serial interval	1 second
Movement threshold	2.0 km/h
Distance unit	km
Speed unit	km/h
Local time	IST (UTC + 5:30)